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Group 3: PM Case Assignment #2
9/29/2025

CSC-580 PM CASE: THIRD AVENUE SOFTWARE HEALTH-CARE APP PROJECT

This case is new for the ninth edition of *Information Technology Project Management*. The case provides an opportunity to apply agile and Scrum principles to project management.

Each part of the case contains several task assignments to help you explore the use of agile and Scrum principles.

Part 2: Project Scope Management

As one of the two senior programmers at Third Avenue, you have been selected to run the project for developing the health-care app. You will be joined by the following colleagues on the project team:

- Eric, a junior programmer who is considered by his peers to be the author of some of the best code at the company. You have also designated Eric to be the project ScrumMaster.
- Lia, another talented young programmer
- Brianna, a marketing representative who has experience in health care from a previous job.
- Jack, the regulatory manager at Third Avenue
- Kendra, the Quality Assurance manager

Remember that project scope management is different in agile projects than in traditional project management. For example, participants in agile projects typically spend less time defining scope in early stages of a project. However, Third Avenue has high hopes for the health-care app and wants to make sure that all team members work out some basic, crucial requirements before proceeding. Also, agile projects generally require more iterations of working software than in traditional project management, so management must be willing to trust the process once the basic requirements are in place and understood.

To help develop scope, agile and Scrum approaches employ *cards*, *user stories*, and *technical stories*. User stories are often written on index cards and then arrayed on a wall or tabletop to help the agile team plan how to implement the ideas into the product. Technical stories are then developed from the user stories. Technical stories can contain one or more technical tasks that developers use to chart progress on a sprint board as work is conducted throughout a sprint. This approach facilitates group discussion, which often leads to a much better set of product specifications than the rather simple ideas expressed on the cards.

One of management's key goals is to have the team develop ideas for completing a minimum viable product (MVP) as soon as possible. An MVP is a streamlined, stripped-down version of a product that can still be released for real-world use and review. It contains a subset of features that will be included in the final version. An MVP must possess several key properties:

- It has sufficient usable features and value that users will buy it.
- These early users will see the potential benefits of the product and trust that it will only improve in later iterations.
- It provides a feedback loop that will help programmers improve the existing features and add new features with minimal delays.

Remember that the overall budget for the project is $350,000, and Third Avenue management would like to see a finished application available in four months. The MVP version, of course, must be available much more quickly—management wants it to be ready to ship in six weeks. The project team has decided that sprints will be done every two weeks, so the MVP version must be ready to ship for use and review after three sprint cycles. The budget for completing the MVP is $120,000.

*• Open a new Microsoft® Word document and complete the **Tasks** below. • Save the file on your computer with your last name in the file name. (Example: part 1 tasks _Jones.doc)*
*• Click the **Choose File** button to select and upload your saved document.*

Tasks

1. Based on what you have learned in Part 2, complete the project charter you began in Part 1.

Updates in Green

Project Title: TBD	
Project Start Date: 9/18/25	**Projected Finish Date:** 1/19/26
Key Schedule Milestones: · **Minimum Viable Product (MVP) available in 6 weeks from Project Start.** · **Complete and deliver final product by January 19th**	
Budget Information: Third Avenue has allocated $350,000 towards the project although the entire scope is not completely defined. To dial in estimates management has had the programmers develop a complexity analysis for each major feature in the app, which in turn may help the estimating process later on in the project. After more deliberation, management decided that a Minimum Viable Product needs to be produced 6 weeks from project start and the allocation for that milestone is $120,000	

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Project Objectives: The objective of the project is to create a mobile app that helps Third Avenue find its corporate identity and establish a permanent footing. The project team aims to deliver a health care tracking app that will:

- Appeal to a more universal market as opposed to the niche markets of the past.
- Create a highly *usable* app to tend to the older target market.
- Effectively monetize the app generating an acceptable ROI.
- Create MVP version of app 6 weeks from project start that:
  - Has sufficient usable features and value for user to purchase.
  - Early users see potential benefit from using the product and trust that it will only improve in later iterations
  - Provides a feedback loop that enables programmers to improve Existing features and add new ones quickly.
- Must complete MVP version to a budget of $120,000
- Produce a Finalized version of the application in 4 months.
- Meet or beat the $350,000 budget allocation target.

Success Criteria: The project will be considered a success if the team delivers a working version of the app within the 4-month deadline and achieves or beats the budget set forth of $350,000. The app must be deemed usable relative to the target audience by management and have a written monetization plan. The app must have an MVP release to customers 6 weeks from project start. The cost for the MVP deliverable must meet the budget goal of $120,000

Approach:

- By 9/19, identify personnel for the project and define roles within a Scrum development team.

- Since Management decision to have MVP 6 weeks from project start, previous approach measures were completed in expedited fashion:

    - Personnel and roles completed: 9/19

    - Scope dialed in more clearly completed 9/19

- First Scrum sprint scheduled to begin 9/22 will last 2 weeks and will aim to begin to lay the framework for the app.

- Second 2-week Scrum Sprint to begin 10/8 after first sprint review and retrospective. MVP will be released at conclusion of 2nd sprint. The team will have 1 week to polish and prepare MVP Deliverable and presentation to management 10/31.

- The 3rd Sprint will be 2 weeks long, begin on 11/3, and will aim to resolve to tackle an updated product backlog including feedback from MVP release.

- The 4th Sprint will be 2 weeks long, begin on 11/19, and will aim to tackle product backlog issues and begin to incorporate changes from MVP app feedback loop.

- The 3rd Sprint will be 2 weeks long, begin on 11/3, and will aim to tackle items from the product backlog including feedback from MVP release.

- The 4th Sprint will be 2 weeks long, begin on 12/8, and will aim to tackle items from the product backlog including feedback from MVP release.

- The 5th Sprint will be 2 weeks long, begin on 12/24, and will aim to tackle items from the product backlog including feedback from MVP release. This Sprint would aim to only tackle the top final remaining backlog items. Work must be monitored closely to budget and scope of the Sprint must be viable in observance of multiple Holidays during this period.

- The project allows for one and a half weeks of “Technical Float”, with the fifth sprint concluding 1/7, during which the product can be tested and polished for the last time. Also, the team can work on the presentation of the software to project sponsors and stakeholders. The team will need to be highly cognizant of hours allocated to the project at this point to help maintain budget goals. Ideally this “technical period” lasts for 1 week and the team delivers the product early to project finish date by 5 days on 1/14.

Roles and Responsibilities			
Name and Signature	***Role***	***Position***	***Contact Information***
Eric	**Scrum Master**	**Junior Programmer**	**555-555-5555**
Lia		**Programmer**	**555-555-5556**
Brianna		**Market Rep**	**555-555-5557**
Jack		**Regulatory Manager**	**555-555-5558**
Kendra		Quality Assurance Manager	555-555-5559
Comments: (Handwritten or typed comments from above stakeholders, if applicable)			

2. Part 1 of this case listed the key features needed for the app. The list is quickly summarized here:
   - A fitness tracker for recording health information, such as blood pressure and cholesterol
   - A medication tracker (electronic pillbox) with a calendar and alarm notifications
   - An electronic address book for recording contact data of doctors and other health care professionals
   - An emergencies list for storing vital phone numbers and addresses to provide quick access to hospitals, urgent care clinics, children, and friends in an emergency. List entries will trigger interactive GPS mapping software to help locate hospitals and other health-care venues.
   - An emergency information list in which customers store important data about themselves in case it is needed in an emergency.

- A resources feature that lists links to other popular online health sites, such as WebMD
- A payment feature that tracks health expenses and allows customers to make related payments through their phones.
- Usability issues

Using this feature list, develop a set of cards, user stories, and technical stories to describe the software requirements for the health-care app. Remember from your course readings that user stories describe what users need to do to execute a task or perform a job function, focusing on the “who,” “what,” and “why” of a requirement in a simple, concise way. ‘’

Use Case Name:	Fitness Tracker
Primary Actor:	User
Description	1. The user logs into the app on their phone. (User)
	2. The app presents the user with options to select that will take them to different parts of the App. (App)
	3. The User selects the option for recording health information. (User)
	4. The App loads the area for recording health information and presents the user with information of previous health information and areas to input new information. (App)
	5. The User inputs new health information. (User)
	6. The app stores, updates, and presents the user with the new inforation. (App)

Use Case Name:	Medication Tracker
Primary Actor:	User
Description	1. The user logs into the app on their phone. (User)
	2. The app presents the user with options to select that will take them to different parts of the App. (App)
	3. The User selects the option for Medication Tracker.
	4. The App loads a calendar that displays each day of the month and the required medications and their time to be taken for each day.(App)
	5. The User inputs may input new meidcations into the calendar or update current ones. (User)
	6. The app stores, updates, and presents the user with the new inforation. (App)
	7. The app sends notifications to phone when medications are due to be taken given the perscribed schedule. (App)

Use Case Name:	Contact Data
Primary Actor:	User
Description	1. The user logs into the app on their phone. (User)
	2. The app presents the user with options to select that will take them to different parts of the App. (App)
	3. The User selects the option for Contact Data. (User)
	4. The App loads and displays a database of all medical contacts and their contact information.(App)
	5. The User inputs may input new contacts and their respective information (User)
	6. The app stores, updates, and presents the user with the new information. (App)
	7. The User can Select any of the contacts information to begin contacting in that form i.e. email, phone, text, etc.. (User)

Use Case Name:	Emergency List
Primary Actor:	User
Description	1. The user logs into the app on their phone. (User)
	2. The app presents the user with options to select that will take them to different parts of the App. (App)
	3. The User selects the option for Emergencies. This option Sticks out from the rest and is highlighted in Red (User)
	4. The App immediately brings up an emergency contact list that the user has set up and maps directions to the nearest Emergency room, the app then asks if the user would like to be connected to 911. (App)
	5. The User inputs can choose to engage or deny 911 services. If denied they can review emergency information and engage the contacts however they choose. They can also add new information. (User)
	6. The app engages selected contacts or stores, updates, and presents the user with the new inforation. (App)

Use Case Name:	Emergency info
Primary Actor:	User
Description	1. The user logs into the app on their phone. (User)
	2. The app presents the user with options to select that will take them to different parts of the App. (App)
	3. The User selects the option for Emergency Information. (User)
	4. The App loads and displays a database of all information the user deems necessary to display such as Allergies, DNR, etc.. (App)
	5. The User inputs may input new emergency information (User)
	6. The app stores, updates, and presents the user with the new information. (App)

Use Case Name:	Web Resources	
Primary Actor:	User	
Description	1. The user logs into the app on their phone. (User)	
	2. The app presents the user with options to select that will take them to different parts of the App. (App)	
	3. The User selects the option for web resources. (User)	
	4. The App brings up a predetermined list of sponsored websites. (App)	
	5. The User chooses a website (User)	
	6. The app directs the user to the webistes home page (App)	

Use Case Name:	Payment Tracker	
Primary Actor:	User	
Description	1. The user logs into the app on their phone. (User)	
	2. The app presents the user with options to select that will take them to different parts of the App. (App)	
	3. The User selects the option for Payment Tracking (User)	
	4. The App brings up a history of transactions from the users bank account that are health related. (App)	
	5. The User can review (User)	
	6. The app updates when a new transaction is made (App)	

3. The "Collecting Requirements" section of Module 5 discusses several methods for gathering requirements, including questionnaires, surveys, stakeholder interviews, prototyping, and context diagrams. Based on your knowledge of agile and Scrum, which of these methods should the Third Avenue team use to collect requirements for the project? Write a two-paragraph response to defend your answer.

- Firstly, I believe we need to prioritize stakeholder interviews and rapid iterative prototyping. Then develop a context diagram using short targeted surveys to validate assumptions. The surveyors should be early adopters and all done during MVP phase. This ties back to Scrum's important point for direct conversation and working increments, enabling "inspect-and-adapt loops" suited to a six week to eight week MVP and two week Sprints increments.
- Secondly, we need to execute structured interviews and workshop conversations to shape user stories with well-thought-out acceptance criteria for backlog refinement. Next, we would need to translate the most salient points (potentially via a Pareto Analysis) into quick prototypes and deployable/shippable increments via each Sprint. Next, this means using Sprint Reviews to obtain real-time feedback, which is also collected with brief surveys. The feedback of this supports our early context diagram, helping clarify system boundaries. The last portion would be to validate that the entire project is aligned with compliance, especially via each interface to constrain/validate the scope and minimize rework, and reducing the risk of scope creep.

4. Develop an initial scope statement. Make sure to follow the detailed process shown in Module 5. Recall that a good scope statement requires some of the items shown in the following table.

Components of a scope statement
Information from the project charter
Product scope description
Functional and design specifications for developing software
Product user acceptance criteria
Detailed information for project deliverables
Project boundaries, constraints, and assumptions
References to supporting documents, such as product specifications or corporate policies

Based on your work in developing the software requirements and scope statement, develop a list of features that will become the MVP for the first iteration of the health-care app. For example, the programmers' initial ideas for the app include (a) an electronic address book for recording contact data of doctors and other health-care professionals; and (b) an emergencies list for storing vital phone numbers and addresses of hospitals and other emergency venues. Should these two items be combined in the MVP version? Consider such issues as you develop your list.

Scope Statement (Sprint 1)

Project Title: Third Avenue Health App MVP **Date:** 09/20/25 **Prepared by:** Eric (Scrum Master)
Project Justification: The purpose of this project is to develop a Minimum Viable Product (MVP) of the Third Avenue health app with the goal of achieving customer acceptance. The MVP will allow users to create and maintain an electronic address book for health-care professionals and an emergency contact list for vital facilities. Development will follow an agile methodology with two-week sprints, a total ship date of six weeks, and a budget of $120,000.

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Product Characteristics and Requirements:

1. Electronic address book for recording contact information of doctor and other healthcare professionals
2. list for storing vital phone numbers and addresses of hospitals and other emergency venues.
3. Application is delivered in 6 weeks and within a budget of $120,000

Product User Acceptance Criteria:

Users will be able to create an electronic address book for recording contact data of doctors and other health-care professionals; and an emergency list for storing vital phone numbers and addresses of hospitals and other emergency venues.

Summary of Project Deliverables

Project management-related deliverables: business case, charter, team contract, scope statement, WBS, schedule, cost baseline, status reports, final project presentation, final project report, lessons-learned report, and any other documents required to manage the project.

Product-related deliverables: research reports, design documents, software code, hardware, etc.

1. Agile Sprint 2 week intervals
2. Lessons learned